

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458601.7879 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.012
Transit depth (Rp/Rs)^2	3.33 +/- 0.44 [%]
Orbital Inclination (inc)	89.93 +/- 0.83
Airmass coefficient 1 (a1)	1.0471 +/- 0.0084
Airmass coefficient 2 (a2)	-0.027 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	#3 - [581, 112]
Optimal Aperture	3.99
Optimal Annulus	13.33
Transit Duration (day)	0.1139 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.182 ± 0.012 vs 0.1326 (deviation 4.12σ)
Duration fit vs published	0.1139 d vs 0.1238 d (deviation 3.52σ)
Mid-time O-C	0.3 min
Depth SNR	15.2
Residual scatter	0.8 %

Light curve

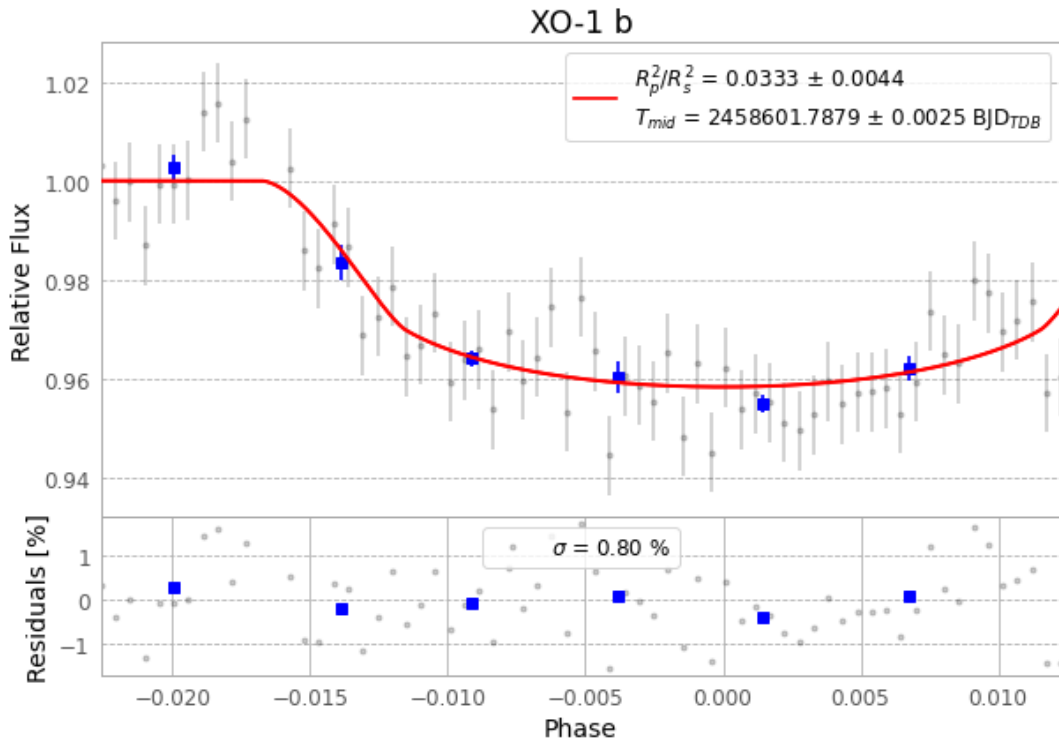


Figure 1 — FinalLightCurve_XO-1 b_2019-04-27.png (SHA-256 0c5d035fa809139c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 73], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1a8011d87da84668...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), XO-1190428043913.FITS → XO-1190428075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-190428.log (0527bc036e3b...), FinalLightCurve_XO-1 b_2019-04-27.png (0c5d035fa809...), FinalLightCurve_XO-1 b_2019-04-27.csv (ca4b4eade544...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)